

Homework 6 - Bayesian Workflows & Model Comparison

Due on May 15th, 2026

In previous weeks, you built Markov Chain Monte Carlo (MCMC) engines from scratch. In modern physics and data science, however, we use automated Probabilistic Programming Languages (PPLs) like PyMC or Stan, which utilize Hamiltonian Monte Carlo (HMC/NUTS). Your job is no longer to write the sampler, but to design the **Bayesian Workflow**. In this assignment, you will build generative physical models, execute prior/posterior predictive checks, diagnose Hamiltonian divergences, and use Leave-One-Out Cross-Validation (LOO-CV) to declare a winning physical model for an aerospace materials dataset.

The Aerospace Dataset: Material Fatigue (S-N Curve)

You are an engineer analyzing the fatigue life of Aluminum Alloy 7075-T6, used in aircraft wings. You subjected 10 identical samples to continuous cyclic stress of varying amplitudes (S , in Megapascals) until they catastrophically fractured. You recorded the number of cycles to failure (N).

Because fatigue life varies by orders of magnitude and is strictly positive, the noise is multiplicative, not additive. Therefore, we model the failure cycles using a **Log-Normal distribution**.

Experimental Fatigue Data

Sample i	1	2	3	4	5	6	7	8	9	10
Stress S_i (MPa)	400	380	350	320	290	260	230	200	180	160
Cycles N_i	1.2e4	2.6e4	6.2e4	1.4e5	4.1e5	1.1e6	3.5e6	1.5e7	5.2e7	1.8e8

Problem 1:

Model 1: Basquin's Equation & The Generative Workflow – Points: 50

The standard physical model for high-cycle fatigue is Basquin's Equation:

$$N = A \cdot S^{-\beta}$$

Taking the natural logarithm of both sides gives a linear relationship in log-space:

$$\ln(N) = \alpha - \beta \ln(S)$$

where $\alpha = \ln(A)$. We assume the observed cycles N_i are Log-Normally distributed around this theoretical mean, with a standard deviation σ . Thus, the generative likelihood is:

$$N_i \sim \text{LogNormal}(\mu_i, \sigma) \quad \text{where} \quad \mu_i = \alpha - \beta \ln(S_i)$$

- Prior Formulation & Predictive Checks:** Use PyMC (or your preferred PPL) to code Model 1. You must choose weakly informative Normal priors for α and β , and a Half-Normal prior for σ . *Task:* Before conditioning on the observed data, simulate 500 draws from your Prior Predictive Distribution. Plot the prior predictive S-N curves on a Log-Log scale. If your priors generate unphysical curves (e.g., curves where life N *increases* with higher stress, meaning $\beta < 0$), you must adjust your priors until they reflect physical reality.
- Inference & HMC Diagnostics:** Condition the model on the provided data and run the NUTS sampler (e.g., 2000 draws, 4 chains). *Task:* Report the Gelman-Rubin statistic ($\hat{R}$) and the Bulk

Effective Sample Size (ESS) for α, β , and σ . Did the sampler report any divergent transitions? Include the Trace Plot to visually prove convergence.

- c) **Posterior Predictive Check (PPC):** *Task:* Generate and plot the Posterior Predictive Distribution. Overlay the actual observed data points (N_i) onto the 95% Credible Interval band of your simulated model predictions. Does Model 1 adequately capture the variance of the data?

Problem 2:

Model 2: The Fatigue Limit (Stromeyer's Equation) – Points: 30

Looking at the data, you might suspect that at very low stresses, the material will never fail. This implies a "Fatigue Limit" or endurance limit (S_e). Stromeyer proposed an updated physical model:

$$N = A \cdot (S - S_e)^{-\beta} \implies \mu_i = \alpha - \beta \ln(S_i - S_e)$$

For this model to be physically valid, S_e must be strictly less than the lowest observed stress (otherwise the logarithm is undefined).

- a) **The New Generative Model:** Code Model 2. Define a bounded prior for S_e such that $0 \leq S_e < 155$ MPa. Keep your priors for α, β , and σ similar to Problem 1.
- b) **Inference:** Run the NUTS sampler. *Warning:* This non-linear parameterization creates a complex posterior geometry (a "funnel"). If you encounter divergences, try increasing your `target_accept` parameter to 0.95 or 0.99 to force the Hamiltonian engine to take smaller time steps.
- c) **Results:** Report the mean and 94% Highest Density Interval (HDI) for the fatigue limit S_e .

Problem 3:

Model Comparison via Information Theory – Points: 20

You now have two competing physical models: Basquin (Model 1) and Stromeyer (Model 2).

- a) **Leave-One-Out Cross-Validation:** Use the `arviz.compare` function (from the `arviz` module of Python) to perform Pareto Smoothed Importance Sampling (PSIS-LOO) on both models.
- b) **The Verdict:** Report the ELPD (Expected Log Predictive Density) difference and its standard error. Based purely on out-of-sample predictive power, which model is mathematically superior? Does the addition of the S_e parameter justify its added complexity, or does it overfit the data? Provide a 3-4 sentence physics-informed conclusion.

Submission Formatting: Submit a single PDF containing your commented Python/PPL code in an appendix, the required Prior/Posterior Predictive plots, the Trace plots, and your written answers/diagnostic tables for Problems 1, 2, and 3.